Embodied Augmentation of Existing Video Exhibits: Reshaping Visitor Engagement Without Replacing the Screen

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Augmenting an existing video exhibit through embodied interaction, this thesis investigates how small physical and digital interventions can reshape visitor engagement without replacing the screen. The study was conducted at a biological museum in Sweden and used a Research through Design approach to develop and test tangible, spatial, and temporal augmentation strategies around an existing natural science video exhibit. The findings show that engagement with screen-based exhibits is shaped not only by the video content, but also by how visitors are invited to approach, touch, wait, enter, and stay with the experience. The book-shaped artefact made the interaction feel more intentional and participatory, while also showing that tangible interaction needs clear invitation and reliable feedback. Spatial changes such as curtains and lighting supported focused attention but also created hesitation when the boundary felt unclear. Temporal cues helped communicate the video's duration, although they worked best when they stayed peripheral rather than visually dominant. Overall, the thesis contributes design insights for lightweight embodied augmentation, showing how existing museum exhibits can be extended through careful attention to invitation, material meaning, spatial openness, and temporal clarity.

CCS CONCEPTS • Human Computer Interaction

Additional Keywords and Phrases: Embodied Interaction, Tangible Interaction, Materiality, RtD, Museum Studies

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1 INTRODUCTION

Screen-based digital displays have become common in public environments, including museums, because they can present dynamic information and be updated more easily than static media and operate at relatively low cost [40]. In museum context video can also support linear narrative, demonstrate processes, show changes over time, and present complex issues [34]. However, these installations can also position visitors mainly as passive viewers [29]. In busy museum environments, visitors frequently move quickly between exhibits, and lengthy videos

compete with surrounding objects, sounds, people and other stimuli for attention [34]. Visitor attention is also shaped by exhibition length, spatial layout, information presentation, lighting, sound, and the overall museum atmosphere [27]. As a result, many screen-based installations experience low dwell time and limited embodied engagement [19, 34].

While immersive technologies such as VR, AR, or large-scale interactive installations offer alternative ways of engaging visitors [28, 38], they often require substantial infrastructural change, technical maintenance, staff expertise, and financial investment. Recent work on mixed reality museum experiences shows the potential of digital augmentation, remote presence, and object manipulation [28]. However, many museums cannot easily replace or fully redesign existing exhibits. Museum planning literature also emphasises that museum development is shaped by financial, spatial, technical, organisational, and stakeholder constraints [4]. Therefore, there is value in exploring lighter forms of augmentation that can improve visitor engagement within existing exhibition conditions [4, 22].

Although screen-based and kiosk-type exhibits remain common ways of communicating complex topics in museums [3, 29], embodied interaction has been shown to support engagement and meaning making by involving visitors' bodies, actions, and situated experiences [11, 19]. Research on visitor attention and museum fatigue also suggests that interactivity, clear navigation, emotional resonance, and experiential design can help visitors stay focused and reduce fatigue [27]. However, much existing research focuses on fully immersive, newly designed, or extensively redesigned installations rather than lightweight augmentations of existing video exhibits [24, 28]. There is still limited empirical investigation of how distinct embodied material strategies, such as temporal framing, tangible interaction, and spatial or atmospheric modulation, can reshape engagement within existing screen-based formats. This gap highlights the need for comparative research on embodied augmentation strategies within natural science museum contexts.

This gap is suitable for an HCI thesis because it is situated within the field of tangible and embodied interaction in HCI, particularly in relation to museum interaction design. This thesis explores lightweight embodied augmentation strategies that operate as additional interaction layers around an existing video exhibit, rather than replacing the screen or redesigning the digital content. The study was conducted in collaboration with a natural science museum in Sweden, focusing on a video exhibit located in a library-like area. The aim is to investigate how material, spatial, and temporal interventions can reshape visitor attraction, attention, behaviour, and participation within the existing exhibit format.

1.1 Research Question

Main Research Question

- How can different embodied interaction strategies augment an existing video exhibit in a natural science museum to reshape visitor engagement without altering the existing exhibit format?

Sub Question

- How different embodiment strategies: temporal embodiment, tangible interaction and spatial/atmospheric changes influence visitor attention, behavior, and participation around the video exhibit?

2 BACKGROUND

2.1 Museum Exhibit Studies in HCI

Museums have become an important context within Human-Computer Interaction because they bring together people, technology, artefacts, learning, and public space. Unlike many personal digital systems, museum technologies are encountered by visitors who may have different ages, interests, motivations, and levels of prior knowledge [13]. Visitors also have limited time and often move through exhibitions selectively [13]. This makes museums useful places to study how interaction happens in real-world, informal, and situated environments.

Early digital museum systems often focused on providing information through multimedia guides, screens, kiosks, and touchscreen interfaces [18, 34]. These technologies made it possible to present text, sound, image, and video in flexible ways [40]. However, they also shaped interaction around looking, reading, pressing, or selecting. This form of interaction can be useful for learning, but it does not always make full use of the body, physical space, material artefacts, or social movement through the exhibition.

More recent museum interaction research has explored hybrid forms of experience, where digital and physical elements work together [3, 26]. Hybrid museum experiences aim to connect digital technology with the physical museum visit rather than replacing it with a virtual experience [25]. Waern and Sundnes Løvlie describe hybrid experiences as designs that use new technologies to augment, expand, or alter the physical experience of visiting the museum [25].

Hybrid museum experiences combine physical museum spaces with digital or interactive elements, but technology is not the main focus. In Hybrid Museum Experiences, the authors argue that technology should support human experience and meaning making during the museum visit, rather than becoming the center of the experience itself [25]. This is important for the present thesis because the project does not use technology to replace the museum exhibit. Instead, technology and material design are used to support visitor engagement with what is already there.

2.2 Museum Shifting Toward Participatory Experiences

Contemporary museums are moving away from treating visitors as just passive recipients of curated knowledge [13, 35]. Instead, they're increasingly adopting participatory models that invite contribution, dialogue, and co-creation [3, 13, 35]. Meaning isn't just something transmitted one-way from the institution to the audience; it's constructed through the visitors' own personal, social, and physical engagement with the museum environment [13]. In this view, visitors aren't just one-time consumers but active, ongoing participants in the museum ecosystem [13, 35]. Their involvement goes way beyond just "consuming" content to include providing feedback, their own interpretations, and even helping to plan and evaluate museum experiences [35]. Despite this participatory turn, many exhibit formats still rely on screen-based interaction, where visitor engagement is often organized around looking, selecting, pressing, or navigating digital content [21].

2.3 Screen-based Exhibits and Passivity

Screen-based video exhibits are common in museums because they can explain complex topics through audiovisual storytelling [34]. They can show processes over time, provide emotional or personal narratives, and communicate information that may be difficult to show through static objects alone. In natural science museums, this can be especially useful for topics such as animal behavior, ecological change, or climate-related processes.

However, videos also depend on visitors' willingness to stay. Watching a video from beginning to end requires time, and museum visitors often make quick decisions about where to stop and what to skip [34]. Therefore, the success of a video exhibit depends not only on accurate or interesting content, but also on whether visitors notice it, understand the experience it offers, and feel motivated to stay [34].

Serrell's distinction between attraction and holding power is useful here [34]. Attraction refers to whether visitors are drawn to an exhibit, while holding power refers to how long they remain engaged. Boisvert and Slez make a similar distinction in science museum research, describing attraction as whether visitors stop, holding power as the time visitors spend, and engagement as attention shown through looking, reading, touching, manipulating, or discussing an exhibit [6]. Their study also suggests that concrete and interactive exhibits can support stronger holding power and engagement than more abstract or passive forms [6].

For video exhibits, this distinction is important because a visitor may look at a screen briefly without watching the full video. Therefore, improving a video exhibit is not only a matter of improving content. It also involves designing the conditions around the video, including how visitors are invited, how time is communicated, and how the space supports attention.

2.4 Embodied, Tangible and Material Interaction

Embodied and tangible interaction research provides a foundation for understanding interaction beyond screen-based input [11, 19, 20]. Instead of treating interaction only as looking at information or pressing buttons, this perspective considers how people act through their bodies, physical objects, and the surrounding environment. Tangible and embodied interaction is concerned with computational objects in the physical world, including body movement, tangible manipulation, physical embodiment of data, and digitally augmented physical space [19, 20].

This perspective is useful in museums because visitors already engage through bodily and situated actions. They walk through exhibitions, stop, turn, look around, sit, touch, hesitate, and leave. Their engagement is shaped not only by the information presented, but also by what the physical environment makes visible, inviting, or uncertain. Hornecker also argues that tangible interaction should focus on interaction rather than only interface, because this allows the system to be understood as part of a larger ecology and specific context [19].

Materiality is important for understanding how interaction is shaped. Physical qualities such as form, texture, scale, visibility, and placement can influence how people understand an artefact and what actions feel possible [12, 15]. In HCI, materiality also extends beyond the physical object itself, as physical and digital elements are often integrated and experienced together in use [15]. Gross, Bardzell and Bardzell further argue that materiality concerns not only physical form, but also how materials carry information, support interaction, and contribute to meaning making [17].

For museum interaction design, this means that engagement is not produced by digital content alone. It is also shaped by the material and spatial conditions around the exhibit. Physical artefacts, lighting, furniture, boundaries, and interactive objects can influence how visitors approach, interpret, and participate in an exhibition.

2.5 Augmentation VS Replacement

Many museums are interested in digital and interactive technologies, but full replacement is not always possible or desirable. Existing exhibits may still fit the museum's identity, budget, and available infrastructure [23]. Therefore, augmentation can be a practical alternative to replacement.

Digital augmentation is often described as adding digital layers to existing exhibits while leaving the original exhibit [3, 9, 10] intact. Lee, Nam and Choi explain that digital augmentation can leave existing exhibits intact while adding text, audio, video, or virtual elements to enrich the exhibit. They also note that this approach is useful when replacing existing exhibits is difficult or undesirable [22]. Their study is especially relevant because it explores embodied and tangible augmentation for existing museum exhibits and shows how augmentation can increase capture rate, holding time, engagement, learning experience, and positive emotion [22].

The present thesis builds on this general direction but focuses on a screen-based video exhibit rather than a physical science exhibit. It asks how embodied augmentation can support visitor engagement when the original exhibit is a video screen. This makes the focus slightly different: instead of adding digital content to explain a physical object, the project adds tangible, spatial, and temporal cues around an existing digital video experience.

2.6 Positioning of This Thesis

Taken together, this background positions the thesis as a study of embodied and tangible augmentation in a museum context. It connects hybrid and participatory museum research with the specific challenge of screen-based video exhibits, where visitors must notice the exhibit, understand the invitation, and commit time to watching. Embodied and tangible interaction, together with materiality, spatial framing, and temporal framing, provide the analytical focus for examining how physical artefacts, space, and time cues can support engagement with an existing video exhibit without replacing it.

3 THEORETICAL FRAMEWORK

3.1 Tangible and Embodied Interaction

Embodied interaction challenges the idea that interaction is only a mental or screen-based process. Instead, it understands interaction as something that happens through bodily action and situated experience [11, 37]. Dourish describes embodied interaction as the creation, manipulation, and sharing of meaning through engaged interaction with artefacts [11]. This is useful for museum contexts because visitors do not only receive information. They walk, stop, turn, sit, touch, open, hesitate, enter, and leave [13, 19, 25].

Embodied interaction provides the broader lens for analyzing how visitors physically encounter the existing video exhibit and the interventions around it. Engagement is therefore not treated only as watching a screen, but as something shaped by the visitor's body, the artefacts, and the surrounding museum environment [11, 13, 19].

Within this broader embodied lens, tangible interaction is given particular importance. Tangible interaction is used to understand how physical objects can mediate digital interaction [19]. Hornecker describes tangible and embodied interaction as involving body movement, tangible manipulation, physical embodiment of data, and digitally augmented physical space [19]. Hornecker and Buur further frame tangible interaction through bodily, material, spatial, and social forms of engagement [20].

Tangible interaction is not only about adding physical action to a digital system. It also concerns how physical form, computing, and space work together. Baskinger and Gross argue that tangible interaction combines form and computing, where physical form can signal functionality, guide action, and support understanding [5]. This means that the form of an artefact can influence how people recognize possible actions and make sense of the interaction.

Here, tangible interaction is used to analyze how physical artefacts, bodily action, digital response, and the surrounding environment work together in shaping engagement.

3.2 Materiality

Materiality is used as an analytical concept to understand how physical and digital elements shaped visitor expectations and behavior. Rather than treating the book, curtain, lighting, display, seating, and surrounding objects as separate decorative elements, they are analyzed as part of the interaction experience.

This concept helps identify how the book, curtain, lighting, display, seating, and surrounding objects shaped visitor expectations and behavior. These elements are not treated as decoration. They are part of how interaction is experienced [15, 17, 20].

3.3 Spatial and Temporal Framing

Spatial framing refers to how the physical setting guides visitor attention and behavior and memory [11, 13]. Curtains, lighting, seating, and boundaries can make an exhibit feel more focused and intentional [13]. However, spatial framing can also create hesitation if visitors are unsure whether a space is open to them [13].

Temporal framing refers to how time is made visible or understandable within the experience [34]. Video exhibits require visitors to commit time, and this can be challenging in a museum where people move between many exhibits [34]. Serrell's work on museum videos shows that attraction and holding power are not the same, because visitors may notice a video without staying to watch it fully [34]. In this thesis, temporal framing is used to analyze the hourglass, countdown, progress bar, and remaining-time display.

Together, spatial and temporal framing help explain how visitors are invited into the experience, how they understand what is happening, and how they decide whether to stay.

3.4 Use of the Framework in the Analysis

The framework is used to analyze visitor engagement as something shaped through physical artefacts, bodily action, material qualities, space, time, and the museum context. Tangible interaction focuses on how artefacts invited, supported, or confused interaction. Materiality focuses on how the qualities of artefacts and surrounding elements shaped expectations and behavior. Spatial framing focuses on the curtain, lighting, seating, and physical boundaries. Temporal framing focuses on how visitors understood duration and time commitment.

4 METHOD AND METHODOLOGY

4.1 Research Approach: Research Through Design

This research aims to understand how embodied strategies can reshape visitor engagement with an existing video exhibit, without replacing the current setup. The study is built around prototypes, but they are not treated as final products. They are used as research tools to explore what happens when material, spatial, and temporal changes are added around the video exhibit [14, 30, 41].

The prototypes were tested in the real museum setting because interaction meaning does not come only from the object itself. It also depends on the place, visitor, surrounding environment, and situation where the interaction happens [11, 36]. Suchman argues that action is situated and depends on the concrete circumstances in which it takes place, rather than only following a fixed plan [36]. Dourish also explains Suchman's view by

showing that plans do not determine action directly, and people respond moment by moment to the situations they are in [11]. For this reason, the prototypes were introduced into the exhibit area and observed in use.

Research Through Design (RtD) [41] was chosen as the main methodological approach because this study produces knowledge through designing, making, testing, and reflecting on prototypes. In RtD, prototypes are not only possible solutions [14, 41]. They also help explore a design problem and understand how a situation can be changed [30]. This fits the aim because the prototypes helped explore how embodied interaction strategies could support visitor engagement with the existing video exhibit.

This process also connects with Schön's idea of reflective practice. Schön argues that professional knowing is often tacit and embedded in action, judgement, and response rather than only fixed rules [33]. Gaver argues that RtD often produces artefacts, accounts of use, and conceptual insights rather than universal theories, while Redström describes RtD as inquiry through practice, making, and judgement [14, 30].

In this study, knowledge developed through cycles of design, testing, observation, reflection, and refinement. Each prototype helped reveal the exhibit situation differently and informed the next design decision.

The study followed an abductive methodological approach. The early phase was exploratory, using staff discussions, observation, literature review, and sketching to understand the existing exhibit and identify design opportunities. These activities led to three design directions: tangible interaction, temporal framing, and spatial/atmospheric change. Later testing and analysis focused on how each strategy shaped visitor attention, behaviour, and participation. In this way, the study developed through an iterative relationship between empirical material, theoretical concepts, and design reflection.

Therefore, this was a design-led and deployment-oriented inquiry. The artefacts were introduced into the museum, tested with participants and visitors, and adjusted based on what happened in the field. The knowledge came not only from making the prototypes, but also from seeing how they worked in everyday museum conditions.

4.2 Research Context

This study was conducted at a natural science museum in Sweden. It focused on an existing screen-based video exhibit located in a library-like area near the children's play area. The exhibit includes an LED screen, a button panel, seating, bookshelves, brochures, and other surrounding materials. Although partly separated from the wider museum space, it remains visually connected to the exhibition area.

The video content changes periodically and presents nature-related topics through documentary-style videos. Therefore, the study does not focus on one specific video, but on the interaction format around the exhibit and how it could be augmented without changing the original content.

In the existing setup, visitors mainly engage by noticing the screen, pressing a button, and watching the video. This made the exhibit suitable for exploring embodied augmentation, as the interaction was mostly screen-based, with limited bodily activation, spatial framing, and temporal cues. The site therefore allowed tangible, temporal, and spatial strategies to be tested while keeping the existing screen-based format.

4.3 Participants

Participants were recruited from different groups depending on the phase of the study. The study involved museum staff, university students, student-staff members, and museum visitors. This mix was useful because staff

contributed museum knowledge, students tested the strategies in structured sessions, and visitors provided feedback from the real museum setting.

The co-design workshop involved seven museum practitioners. Strategy-specific prototype testing involved fifteen university students from human computer interaction, biological science, and teaching studies. Each participant tested one of the three strategies in the real exhibit setting. The integrated prototype was tested with four museum staff members across two iterations, and quick verbal feedback was also collected from four museum visitors who interacted with the revised setup. Visitor participation was voluntary and depended on their availability and willingness to engage during museum opening hours.

4.4 Data Collection Methods

Data was collected through observation, participatory feedback, co-design workshop activities, prototype testing, interviews, feedback forms, and design documentation. These methods were used together because the study focused not only on what participants said, but also on what they did during interaction with the exhibit.

4.4.1Familiarization and Structured Observations of The Existing Exhibit

The first part of data collection was familiarization with the museum context. This included regular visits to the museum and formal and informal conversations with museum staff, visitors, and other stakeholders. These conversations helped to understand the existing exhibit, visitor needs, staff expectations, practical constraints, and possible opportunities for redesign.

A two-hour structured non-participant observation [8] was conducted in the existing exhibit setting to understand how visitors engaged with the current video exhibit without interrupting their natural behavior. Observation focused on visitor dwell time, entry and exit behavior, body posture and orientation, social interaction around the exhibit, and moments of non-engagement. Visitors who passed the exhibit without stopping or entering were also recorded as important data. In this study, non-engagement was not treated as absence of data, but as part of understanding how the existing exhibit attracted or failed to attract visitor attention.

4.4.2 Participatory Feedback Session

A lightweight participatory feedback activity was conducted to complement the observation data. The setup included an instant feedback board, optional written feedback cards with open-ended prompts, and opportunities for brief verbal comments. In the first iteration, these materials were placed within the exhibit area for one hour, and visitors were invited to participate voluntarily. This activity was conducted in two iterations because the first setup received limited participation. The second iteration was adjusted to make participation easier and more visible and lasted for two hours.

4.4.3Ideation and emergence of design strategies

The three strategies were not fixed at the beginning of the study. This stage was mainly inductive, as data from familiarization (appendix 12), observation (appendix 2), participatory feedback, staff discussions (appendix 11), and design reflections were analyzed to identify repeated concerns and redesign opportunities. A design diary with ideation notes and sketches (appendix 13) were also used to explore possible ways of augmenting the exhibit. These ideas were compared, summarized, and grouped according to similarity (appendix 14).

At the same time, relevant literature helped guide and refine the grouping. Serrell's discussion of video exhibits [34] supported the development of the temporal strategy, as several sketches focused on video duration, time commitment, and ways of making time more visible. Literature on tangible interaction [19, 20] supported the tangible strategy, as several sketches explored physical artefacts, bodily action, and visitor participation.

Observation and staff discussions also pointed to the importance of spatial and atmospheric conditions, such as visibility, focus, and separation from the surrounding space. These ideas were further supported by tangible and embodied interaction literature [11, 19], which emphasizes how interaction is shaped by physical and spatial context.

Through this process, the design ideas were gradually organized into three main strategies: tangible interaction, temporal framing, and spatial/atmospheric change. These strategies therefore emerged from repeated patterns across empirical material, sketching, design reflection, and literature, rather than being fixed from the start.

4.4.4Co-Design Workshop with Museum Staff

From this stage, the study became more deductively oriented, as the three emerging strategies were used to structure the workshop and later prototype development. However, the process remained exploratory, as participants' responses were used to refine the strategies and identify new design considerations. The co-design workshop was conducted with seven museum practitioners for around 45 minutes to explore and refine possible embodied augmentation strategies. At the beginning of the workshop, participants were asked to write down current issues they identified in the exhibit. Then, design probes were introduced one at a time to support discussion and idea generation.

The probes included an hourglass for temporal framing, lighting interventions for spatial framing, and material artefacts with different sizes, weights, and textures for tangible interaction. These probes helped participants respond not only through verbal discussion, but also through handling and reflecting on materials.

The workshop gave museum staff a voice as stakeholders. Their experience with visitors, their understanding of the museum's values, and their knowledge of practical limitations helped shape the later prototype directions. Data from the workshop included sticky notes, discussion notes, and video recording for later analysis.

4.4.5 Strategy Specific Prototype Testing

Each of the three strategies was tested separately with five participants in the real exhibit setting The purpose was to understand how each strategy influenced visitor attention, behavior, interpretation, and engagement before combining them into one integrated setup.

Each session started with a brief introduction and consent process. Participants then interacted with the exhibit while being observed for approximately five to seven minutes. Video recording and photographs were used when necessary and when consent was given. After the interaction, a semi-structured interview was conducted and audio recorded. Finally, participants completed a short feedback form about their experience.

The sessions lasted between 20 and 40 minutes depending on the strategy and the participants' responses. All tests were conducted during museum opening hours, so the prototypes were tested within the real atmosphere of the museum.

4.4.6 Integration Testing

After the separate strategy tests, the three strategies were combined into an integrated prototype. The first version included the book artefact, a small digital display inside the book, curtain-based spatial framing, and lighting changes. This version was tested with two museum staff members. The revised version based on previous test feedback was then tested with two additional museum staff members. Each session lasted around 30 minutes. Quick verbal feedback about the prototype and experience was also collected from four museum visitors.

4.4.7Design Documentation

Throughout the process, design documentation was collected through sketches, prototype notes, photographs, technical testing notes, and design reflections. This documentation helped track how the design changed over time and why certain decisions were made. It was also important for understanding the iterative nature of the Research Through Design process.

4.5 Data Analysis

The interview and written qualitative data were analyzed using thematic analysis, which Braun and Clarke describe as a flexible method for identifying, analyzing and reporting patterns within qualitative data [7]. The material included observation notes, workshop data, interview recordings, written feedback, conversational notes, design reflections, and strategy-specific feedback forms. However, the interviews formed the main basis for theme development, while observation notes, written feedback, and feedback forms were used to support, compare, and refine the analysis.

First, the audio recordings were transcribed locally using the built-in transcription function on the researcher's smartphone in offline mode. The recordings were then reviewed again, and the transcripts were corrected manually where needed. This helped the researcher become familiar with the data and notice repeated ideas, comments, and behaviors.

After familiarization, meaningful phrases from what participants said during interaction with the prototypes were coded. The initial codes were then reviewed and revised. Similar codes were grouped into clusters, and these clusters were developed into candidate themes.

The analysis was informed by the study's focus on three embodied strategies: tangible interaction, temporal framing, and spatial/atmospheric change. This was because the interview prompts, observation focus, and feedback forms were structured around these strategies. At the same time, the analysis remained open to unexpected findings, such as confusion, hesitation, surprise, distraction, and participant suggestions.

Alongside thematic analysis, design reflections were also analyzed as part of the Research Through Design process. This included looking at prototype changes, technical limitations, material decisions, and moments where the design did not work as expected. These reflections helped connect participant feedback with concrete design decisions.

The analysis moved between three kinds of material: what happened during the interaction, what participants said about the experience, and what this suggested for design. This helped connect observation, participant feedback, and design reflection The aim was not statistical generalization, but the generation of design knowledge about how embodied augmentation strategies can reshape visitor engagement in a museum context.

4.6 Ethical Considerations

This study was conducted in collaboration with museum staff, and the prototype interventions were supervised to prevent damage, misuse, or safety issues in the museum environment. Since the study took place in a public museum setting, ethical care was important throughout the process.

Participants in the co-design workshop and prototype testing received information about the study before taking part. They were informed that they could stop at any time. Printed consent forms were signed by workshop participants, student participants, student-staff members, and museum staff who took part in the testing sessions.

Visitor participation was handled informally because the visitor feedback was brief and voluntary. Visitors were only invited to comment or give feedback if they were willingly engaged with the setup. No identifiable personal information was collected from visitors. Observation focused on behavior around the exhibit, such as entering, stopping, watching, leaving, or interacting with the prototype, rather than identifying individual visitors.

Audio recordings, video recordings, photographs, notes, and feedback forms were stored securely. Audio data was transcribed using the built-in transcription feature on the researcher's phone, without uploading the data to external platforms. The data was anonymized during analysis, and participant names were not used in the thesis.

The temporary installations were also discussed with museum staff to make sure they were suitable for the space. The curtain setup, lighting changes, and physical artefacts were arranged with attention to visitor safety, movement, and the existing museum environment.

5 RESULTS AND ANALYSIS

5.1 Prototype development

After the three design strategies had been identified and discussed with museum staff, the project moved into iterative prototyping. The purpose of this phase was to translate the emerging ideas into testable forms and to explore both technical feasibility and material expression.

5.1.1Tangible Interaction Prototype

This study explored a tangible interaction that allowed visitors to trigger video playback through a physical artefact. Instead of pressing a button or interacting with the screen directly, visitors interacted with a book-like object placed near the display. During the staff workshop, participants discussed replacing the existing button-based interaction with a physical artefact that could create an embodied commitment before the video started. Several artefact ideas were considered, including levers and objects that visitors would need to hold during the video, but these were seen as potentially distracting from the video content. The book-like form was selected because it fitted the library-like setting of the exhibit and offered a familiar opening action. During the same workshop, one participant connected the idea to the action of opening a book, saying, "you open a book and all of a sudden switch on the screen". This supported the decision to use an artefact that could invite interaction at the beginning of the experience without demanding attention throughout the video itself. The prototype design and testing setup are shown in Figures 1 and 2. Further technical development, including the hall-effect sensor and magnet mechanism [1], Arduino and Tinkercad testing [2, 39], GPIO compatibility issues [16], and the later ESP32/ESP-NOW wireless setup [31, 32], is provided in Appendix 8. For testing purposes, only one of the four videos was triggered by the artefact. During the strategy testing, the buttons were not covered, giving participants the freedom to engage with both the existing buttons and the artefact.

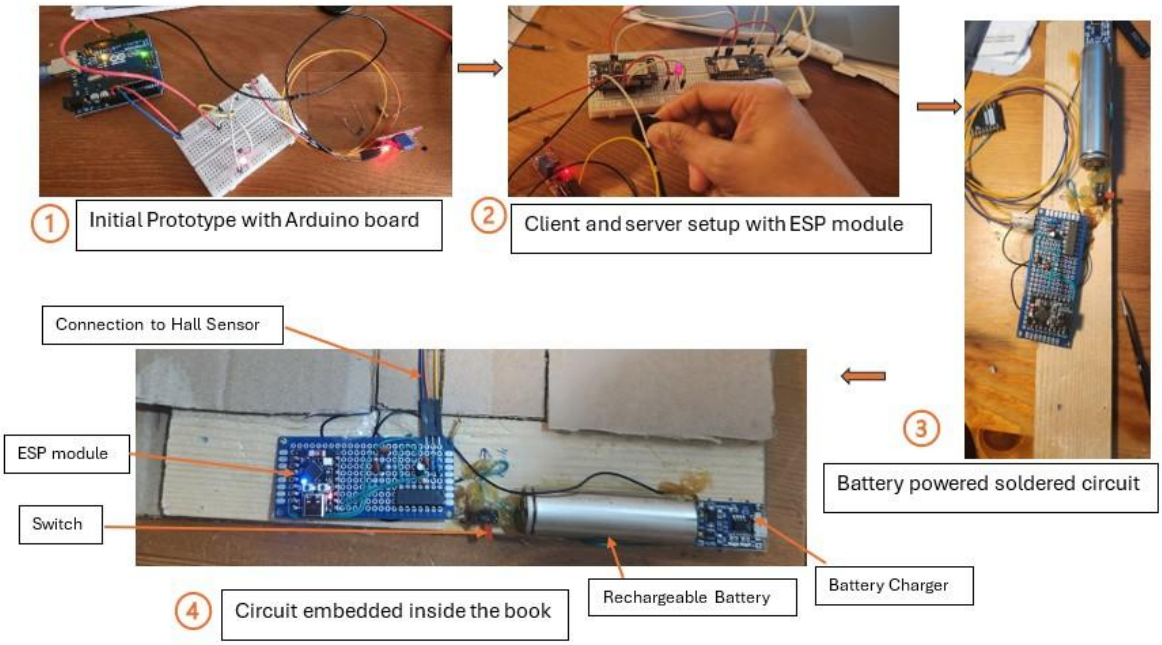


Figure 1 : Iterative development of tangible interaction

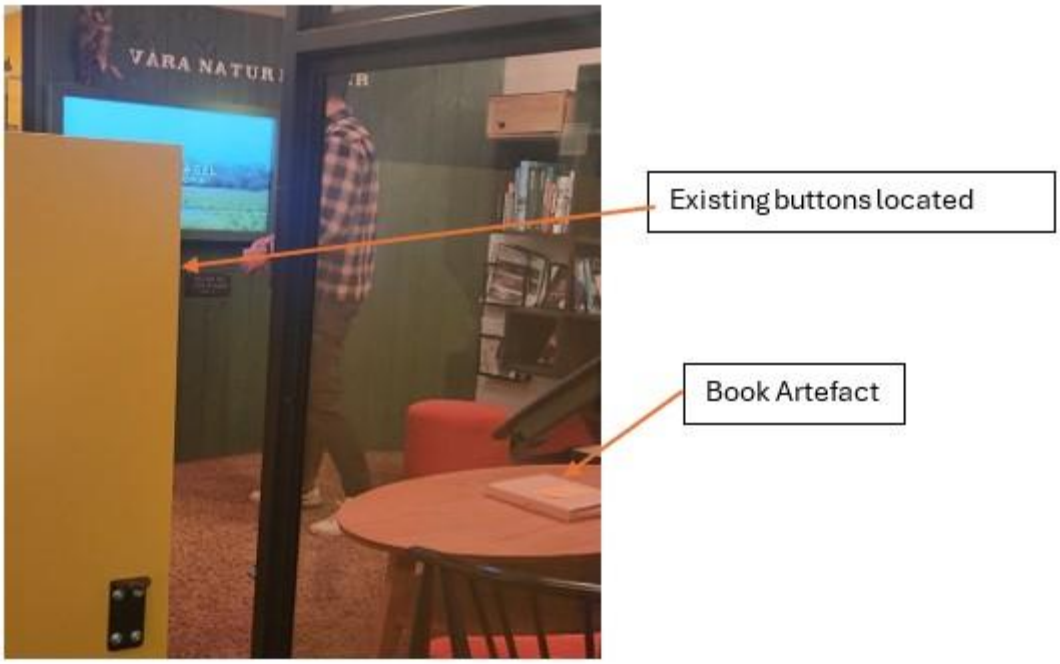


Figure 2: Tangible interaction test setup

5.1.2Temporal Framing Prototype

The temporal framing strategy used an hourglass as a physical representation of time. During the staff workshop, participants discussed the hourglass concept and possible alternatives. One participant suggested that an hourglass could relate time to the video in a more material way than simply showing a number or duration, saying that an "*hourglass is a way of relating time to the video*". However, another participant pointed out a possible limitation, noting that "*an hourglass might attract attention away from the video*". These comments helped motivate the use of an hourglass to explore whether a physical time cue could make the video duration more visible and support visitors' attention and willingness to stay.

The hourglass was approximately 135 mm tall, contained blue sand, and was calibrated for around three minutes. It was placed on a small rack on the right side of the existing button panel. The hourglass was also highlighted using the built-in spotlight in the exhibit area, so that it became more noticeable as part of the interaction setup (see figure 3).

This strategy did not use any specific digital technology. Instead, it explored how a simple physical object could create temporal awareness around the video exhibit.



Figure 3 : Hourglass setup

5.1.3Spatial and Atmospheric Prototype

The spatial and atmospheric strategy focused on changing how the video exhibit was framed within the museum space. In the existing setting, the video exhibit was partly separated by two book racks, but it was still visually open to the surrounding area. During the staff workshop, participants discussed possible ways to make the exhibit feel more focused and intentional, including curtains, "*We could have a curtain system*", lighting, "*We have a new lighting system [...] We used blue light for the ocean topic, and it was effective*", and spatial separation. Based on this, a temporary spatial boundary was created using curtains.

A temporary structure was created to hang the curtains around the exhibit area. The curtain material was selected after discussion with the museum technician. The selection considered how well the color matched the existing space, the thickness of the material, its ability to reduce noise disturbance, availability, and practicality for temporary installation.

Lighting was also used as part of the atmospheric change. When the video started, the light was dimmed, and when the video finished, the lighting returned to its previous state. This helped make the beginning and ending of the video more noticeable in the space (see figure 4).

Visitor safety was also considered during the setup. The curtain placement and hanging structure were arranged carefully to avoid accidents and to make sure the temporary intervention did not block movement or create risks in the museum environment. This prototype explored how spatial enclosure, softer boundaries, and changes in atmosphere could influence visitors' attention and willingness to enter the exhibit.

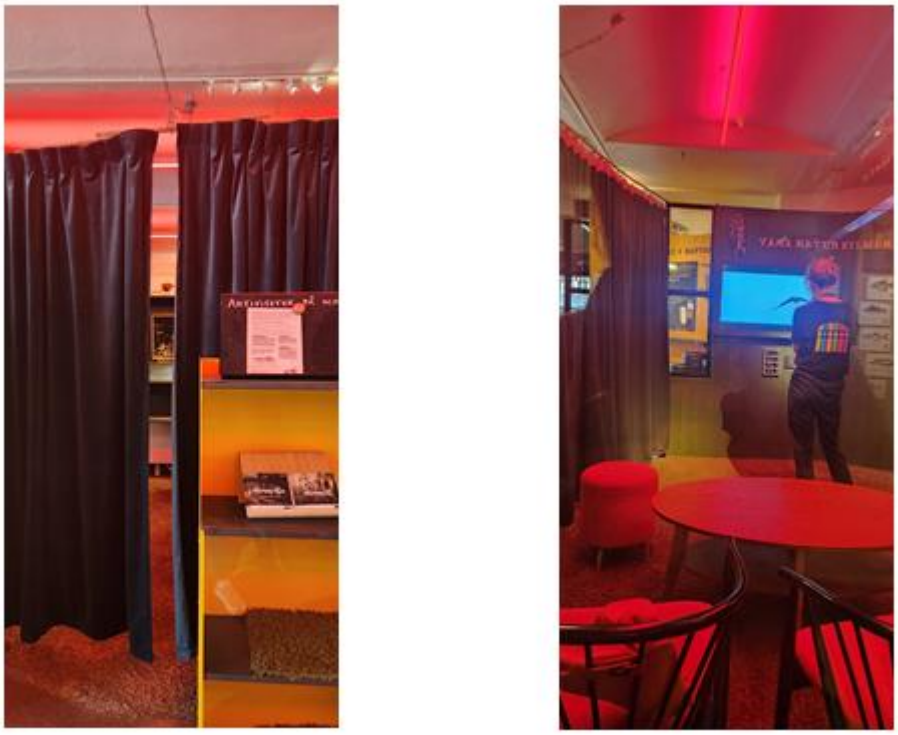


Figure 4 : Spatial changes setup

5.1.4Integrated Prototype: Iteration 1

After the three strategies were tested separately, they were combined into one integrated prototype. The strategy tests suggested that each strategy supported different parts of engagement: the book invited action, the curtains and lighting supported focus, and the temporal strategy made the video duration more visible. Therefore, the integrated prototype explored how these elements could work together as one experience.

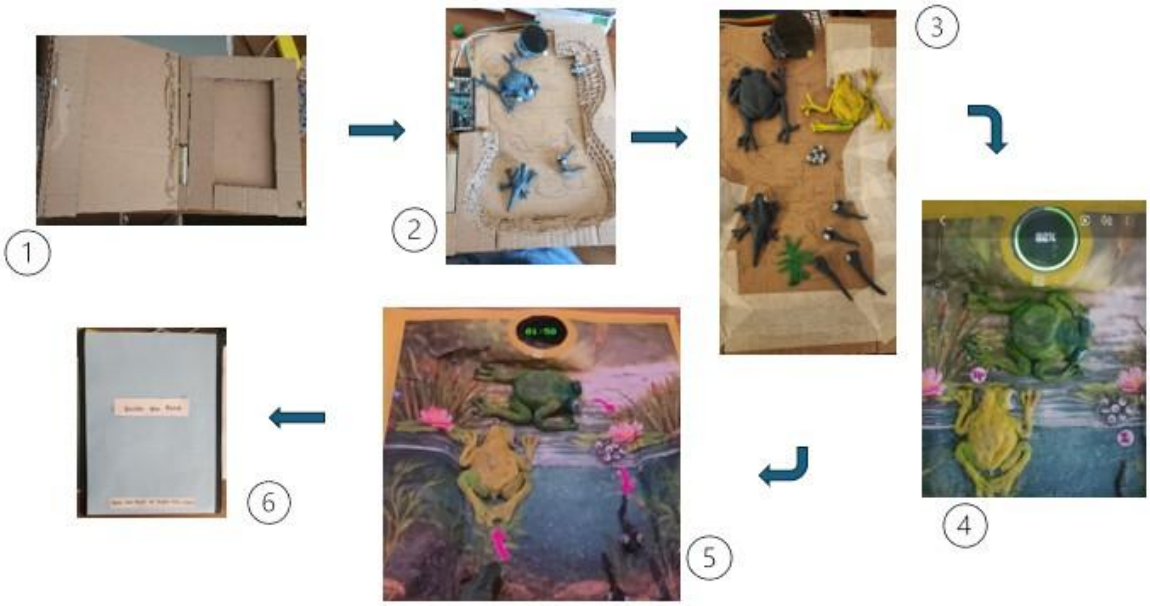
The display was added because observation and participant feedback showed a need for clearer feedback after opening the book. Some participants seemed unsure whether the action had worked or whether the video had

started. One participant suggested that "*when I open the book, maybe the speakers start*," while another noticed that "*this thing started very quickly*". To make the sequence clearer, the display showed the following messages:

Welcome → Video starting, stand by→Countdown 3, 2, 1, 0 →Look at the main screen→Progress bar →Complete

The physical hourglass was replaced by a progress bar on the small display. This decision was informed by feedback from the temporal framing test, where one participant suggested that "*not hourglass, but time becomes visible*" Another participant suggested that "*a progress bar on the video*" might be more helpful than only using the hourglass. Since this study did not change the video content or interface, the idea was adapted into a progress bar on the small display instead. The empty compartment inside the book was also developed into a small pond diorama to connect the artefact more closely with the video content. This followed feedback that the book could include "*short snippets of information*" to guide visitors toward the video topic. The book cover was changed to "Inside the Pond," with the prompt "Open the book to begin the video" (see figure 5).

The book was placed on a table in front of the screen, and the existing buttons were covered so that the book became the main trigger. The same curtain setup from the spatial strategy was used, and red lighting was added. The lights dimmed when the video started and returned to their previous state when it finished. This first integrated version tested how the strategies could work together to invite visitors in, guide the start of interaction, support focus, and make the viewing sequence feel more intentional.



1. Artefact from strategy testing
2. Design progress 1
3. Design progress 2
4. Diorama with progress bar and misleading numbers
5. Number replaced with arrows and remaining time instead progress bar
6. Book cover

Figure 5: Artefact design iteration in integrated testing

5.1.5Integrated Prototype: Iteration 2

Based on feedback from the first integration test, the second integrated prototype revised the display, internal book labels, countdown timing, lighting color, and book topic. Further details on these iteration changes are provided in Appendix 10.

5.2 Prototype design Reflection

Across the prototype development process, each iteration was shaped by reflections from the previous one. These reflections included participant reactions, technical limitations, material qualities, and how clearly each prototype communicated its purpose. The annotated prototype image (see figure 6) summarizes selected design reflections from the tangible, temporal, spatial, and integrated prototypes. It shows how small design details shaped interpretation, such as whether the artefact felt like a book or a box, whether the timer supported or distracted attention, and how lighting and prompts influenced the overall experience. These reflections helped connect earlier prototype observations to later design decisions.

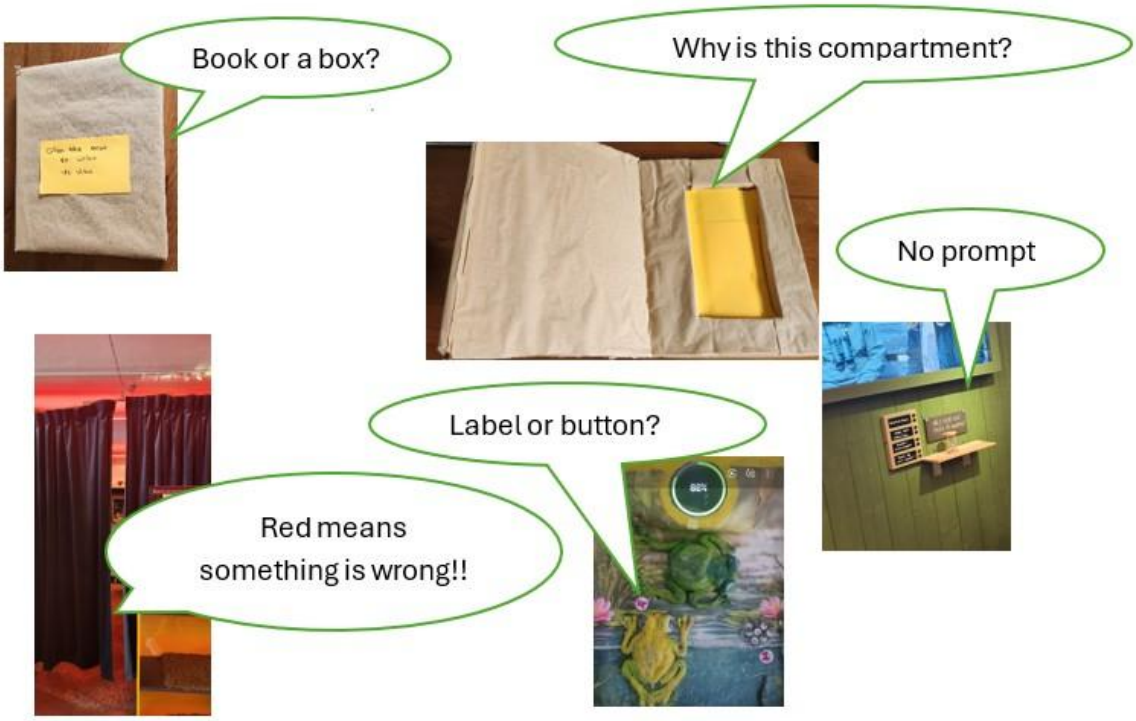


Figure 6: Annotated prototype design reflections across the tangible, temporal, spatial, and integrated prototype iterations.

5.3 Tangible Interaction Strategy

5.3.1Theme 1: From passive watching to embodied participation

The book changed the visitor's role from passive viewer to active participant. Participants who first used the existing buttons often explored the book afterwards, but participants who opened the book first did not usually return to the buttons. This suggests that the book became a stronger entry point. One participant explained that "*the physical commitment of opening it*" felt like a "*more powerful gesture than simply pushing the button.*" Another said that the book was "*a very cool interaction*" because it "*captures the attention*" more strongly. A third connected this to bodily involvement, explaining that the interaction "*involves body and body motions*" and is "*not just about seeing or listening*." Together, these responses suggest that opening the book felt more intentional and meaningful than pressing a button.

5.3.2Theme 2: Surprise, novelty, and memorability

The book also created surprise when participants realized that opening it triggered the video. One participant described the interaction as "*magical*," while another responded with "*Wow*." This surprise was connected to the contrast between a familiar bodily action and an unexpected digital response. As one participant explained, "*I think it was a bit more mysterious to open the book and then try to figure out what the video was trying to tell me than to press the butto*n." Another connected this novelty to the action itself, saying, "*I've never started a video before by opening a book*." This made the interaction more memorable and inviting. One participant added, "*I*

wouldn't want to press more buttons, but I would want to [repeat] that feeling of magic". Opening a book is familiar action, but in this setup, it became a surprising way to activate the video exhibit.

5.3.3Theme 3: Visibility, prompt, and interaction clarity

Although the book shape suggested opening, participants were sometimes unsure whether they were allowed to touch it. This matters in a museum context, where visitors may hesitate to handle objects without a clear invitation. Technical failures also affected clarity. When the book did not trigger the video as expected, one participant said, "*I opened the book and nothing happened, which is a little question mark*," while another noted that "*if it malfunctions, then it distracts".* This shows that tangible interaction needs immediate and reliable feedback. Participants also suggested making the invitation more visible, for example by placing the instruction "*in our viewpoint".* At the same time, the written prompt did help some participants, as one explained, "*I was curious because in the book it was written, open the book to start the video*".

5.3.4Theme 4: Material fit between book, content, and meaning

Participants liked the idea of using a book. One participant said, "*I think it is very clever and smart to choose a book*" while another explained that the book made sense because "*there are multiple stories*" and they "*wouldn't want to touch multiple objects*" while sitting. At the same time, the low-fidelity material quality created some confusion. Some participants felt that the artefact acted more like a box than a real book. The empty compartment inside also distracted participants and created unintended interpretations, such as whether it was a place to put a phone. One participant suggested that the book should include content that connects more directly to the video, such as "*pictures or something like that [to] resonate with the video*".

Overall, the book worked as a stronger entry point than the existing buttons, but it also showed the need for clearer touch permission, reliable feedback, and stronger material connection to the video content.

5.4 Temporal Framing Strategy

5.4.1Theme 1: The hourglass made video duration visible

Participants often connected the hourglass with time because of its familiar form. Some tried to relate it to the video duration like "*I would use it to time the video*", and one participant said the size helped them understand that the video was "*not like an hour*" but "*like a few minutes*". This suggests that the hourglass made time more material and visible, instead of keeping duration hidden inside the screen-based experience.

5.4.2Theme 2: Temporal awareness did not guarantee interaction clarity

Although participants understood that the hourglass represented time, they did not always understand what to do with it. Some did not notice it, while others were unsure whether to touch it, turn it, or simply observe it. One participant explained, "*I usually don't touch things in the exhibition when it's presented there and there's no sign or anything*". Others described its purpose as "*unsure*" and said it had "*no clear connection to anything else*". This shows that temporal awareness alone was not enough. The hourglass needed a clearer prompt and stronger connection to the video.

5.4.3Theme 3: The hourglass competed with the video for attention

For some participants, the hourglass created an additional focus point. One participant said, "*I felt it was more distracting*" while another explained that it might "*hinder my learning*" because there was too much to focus on at the same time. The mismatch between the hourglass duration and the actual video length also caused confusion, as participants began checking whether the two timeframes matched. However, this was not the same for everyone, as one participant said, "*it was not disturbing or distracting me*".

5.4.4Theme 4: Material and sensory integration shaped how fitting it felt

Participants had mixed responses to the hourglass's material fit. Some felt it suited the exhibit because it was physical and subtle. One said it "*did not distract me*" and did not create "*any noise or any light change*". Another connected it to the museum setting, explaining that "*the whole museum had animals and wooden*" materials, so the wooden structure "*made a relation with the nature aspect*". However, others felt it was unnecessary or unnatural. Participants suggested making it larger, such as "*an even bigger hourglass,"* combining it with lighting because "*maybe the light also changes a bit,"* or showing time through a more integrated visual cue. This suggests that the issue was not only whether time should be shown, but how it was materially presented within the space.

Overall, the temporal framing strategy showed that making video duration visible can support visitors' sense of time. However, the hourglass needed a clearer prompt and sometimes competed with the video for attention, suggesting that temporal cues should be clear, accurate, and peripheral.

5.5 Spatial Framing Strategy

5.5.1Theme 1: Spatial framing created a focused, calm, cinema-like enclave

Curtains, seating, and lighting together made the exhibit feel like a separate, calmer zone for watching. One participant said the curtain created "*a little bit of space*" separated from "*all the chaos happening outside*." Some described the space as cozy, theatre-like, or "*kind of cozy, like a little cinema*". Others said the lighting helped them focus, explaining that "*when you press on the button, the lights changed and it was very centered around the films*". This suggests that the spatial changes supported attention by framing the video as something to stay with, rather than quickly pass by.

5.5.2Theme 2: Curtains shaped both curiosity and hesitation

Participant responses suggest that the curtain was a noticeable part of this strategy. For some, it created curiosity and made them want to see inside. One participant said, "*I knew something interesting is inside*," while another said they would feel curious about "*what's in there".* However, others felt unsure whether they were allowed to enter. One participant said the curtain could make the space feel "*more closed off to the public*". This shows that the curtain worked as both an invitation and a barrier.

5.5.3Theme 3: Environmental qualities shaped comfort and staying potential

Participants also commented on furniture, lighting color, seating, sound, and other spatial details. One participant said, "*the chairs are not inviting*" and seemed designed for "*a short time*". Lighting also shaped the atmosphere. Some felt that the setup clearly became "*a space just for that specific purpose,"* while another described seats, dark green colors, and atmosphere as "*cozy*". However, responses to the red lighting were mixed. Some liked it, while

another felt it was "*a bit too much*" and suggested softer colors, such as blue, light purple, or yellow-golden. This shows that spatial framing depends not only on enclosure, but also on comfort and sensory fit.

5.5.4Theme 4: The whole space became part of the interaction

Participants did not experience the strategy only through the curtain and lighting. They also noticed the screen, QR code, animal sculptures, brochures, button placement, sound quality, and seating. One participant described wanting to "*explore a little bit*," read the available information, and "*maybe scan some QR codes*". This suggests that the video exhibit was experienced as a whole environment, not as a separate screen. As one participant explained, the setup made it easier to "*categorize this information*" and "*take things slowly and easily*." The screen's meaning and use were therefore shaped by the physical context around it.

Overall, the spatial framing strategy showed that curtains, seating, and lighting could make the video exhibit feel more focused, calm, and intentional. However, entrance needed to remain clearly inviting, suggesting that spatial framing should balance enclosure with openness.

5.6 Comparison Between Three Strategies

Table 1 : Comparison between three strategies

Comparison theme	Artefact	Spatial	Hourglass
Entry into interaction	Strong: book invited action clearly	Moderate: space framed attention but did not itself trigger action	Weak: object was noticed by some, but action was often unclear
Clarity	Very strong when the book worked reliably	Moderate	Weak
Naturalness	Very strong	Not central, but generally positive	Weak / mixed
Attention and focus	Moderate	Strongest	Weak / mixed
Engagement / involvement	Strong	Strong	Weak / mixed
Temporal awareness	Low	Low	Strongest
Museum fit	Strong	Strong	Mixed

The comparison indicates that the three strategies contributed to different aspects of visitor engagement rather than serving the same function. This comparison was developed by comparing findings from the three strategy tests. Feedback form responses were used as the main source for identifying general response patterns (see Appendix 7), while observation notes and interview comments were used to explain why each strategy was interpreted as strong, moderate, weak, or mixed. Therefore, the table summarizes the combined empirical evidence.

Overall, the findings suggest that embodied augmentation was most effective when it gave visitors clear material action, such as opening the book, or reshaping the surrounding conditions for attention, such as curtains

and lighting. The hourglass showed that time awareness can be useful, but it needs to be more clearly connected to the video and less distracting.

5.7 Integrated Design Testing and Key Findings

Staff feedback showed that the integrated setup felt more coherent than the separate strategies. One participant commented, *"I'll get elevated. The whole setup feels like one coherent thing*". The book remained the strongest invitation because it stood out visually and invited opening. This was reflected in comments such as, "*I did that sitting down and because the book was there on the table, that drew my attention, so I went to it. And when I sat down, I could read that I should open it, but I would have done so anyway*," and "*It seems like a book saying open me*". Staff also described the setup as more cinematic, with one participant asking, "*For me, it's a bit more cinematic, can you see that?*". This suggests that the combination of book, curtain, lighting, and display sequence framed the exhibit as a more complete viewing experience, rather than only a screen with buttons.

At the same time, the first iteration created some misunderstandings. The button-shaped labels inside the book made participants expect to press something, as one said, "*I got the feeling of them being like a button*". The progress bar on the small display also became distracting. In addition, the long gap between the messages and countdown made participants less aware that the video had started because they had opened the book. One participant suggested that "*maybe that time could be a bit shorter*". Based on this feedback, the labels were replaced with arrows, following the suggestion that "*you could replace it with arrows*". The progress bar was replaced with a remaining-time timer, and the timing between the messages and countdown was adjusted. The red lighting was also changed to warmer yellowish lighting to better fit the museum atmosphere.

Visitor feedback on the revised version showed that the integrated setup made the exhibit feel calmer, more private, and more engaging. Visitors described it as "*more private*" and said, "*small change but it changed my experience a lot*". Visitors responded positively to how the curtain, book, and lighting worked together. The curtain created curiosity and a clearer boundary, while the book was more inviting than pressing buttons. One visitor said, "*I prefer this more than button*". The lighting change also helped make the video feel more intentional and focused, as reflected in the comment that the "*light changed to focus more on the video*".

However, both staff and visitors often noticed the remaining-time display only at the end of the video or after a prompt. This suggests that temporal support remained useful but peripheral. It did not need to dominate the experience but could still support visitors who wanted to know how long the video would continue.

Overall, the integrated testing showed that lightweight embodied augmentation can reshape visitor engagement without replacing the screen or changing the video content. The strongest design direction was not to add many separate elements, but to make them work together: inviting visitors in, giving them a clear physical action, supporting focus during the video, and avoiding unnecessary confusion or distraction.

6 DISCUSSION

6.1 Embodied augmentation without replacing the existing exhibit

During this study, the original exhibit was not completely redesigned. The screen, video format, seating area, and general exhibit setting remained mostly the same. The main change was the addition of embodied layers around the video: a tangible artefact, spatial framing, lighting, and temporal support.

This shows that engagement can be reshaped not only by changing the digital content, but also by changing how visitors approach, enter, start, and stay with the video. This connects with hybrid museum experience research, where digital and physical elements are integrated within the museum visit rather than treated as separate experiences [25]. Hybrid museum experiences also augment or alter the physical museum visit through technology while keeping visitor meaning-making central [25].

In this study, the book, curtain, lighting, and timer worked together as a coherent redesign around the existing screen. The video content could still change over time, as it normally does in the museum, while the interaction structure could remain similar. However, if a book-shaped artefact is used, the cover and inner content may need to change according to the video topic to keep the material connection meaningful, as suggested by feedback on material fit and topic connection. Further empirical evidence would be needed to evaluate whether the artefact form fits other museum contexts.

6.2 Visitor engagement as a sequence

The initial observation, staff discussions, and visitor interactions showed that engagement with the existing video exhibit was not only about whether the video content was interesting. It was also shaped by whether visitors noticed the exhibit, understood they could interact with it, and felt invited to stay. Some visitors passed by without noticing the exhibit. Others entered the space but did not fully engage with the video (See Appendix 2).

This suggests that visitor engagement should be understood as a sequence: noticing, entering, understanding what to do, starting the video, watching, and deciding whether to stay. Falk and Dierking's contextual model is useful here because it understands the museum experience through the interaction of personal, sociocultural, and physical contexts, rather than through content alone [13]. They also argue that the museum experience should not be treated as a simple linear process because contexts overlap and merge during the visit [13].

The early participatory feedback activity also gave useful learning. The first version did not collect much meaningful feedback because visitors were in a low-engagement mode. Some accepted the invitation but did not respond, some used cards for children's drawings, and some took empty feedback notes. Although this did not give the intended result, it showed something important about visitor behavior. In a busy museum setting, open feedback materials may not be enough. This learning shaped the later integration testing, where visitors were approached more directly and asked for short verbal feedback with consent.

6.3 Different roles of tangible, spatial, and temporal strategies

The three strategies affected engagement in different ways. The tangible interaction strategy mainly changed how visitors initiated the video. Opening the book created curiosity, surprise, and a more active start than pressing a button. This physical commitment made the interaction feel more intentional. This relates to embodied and tangible interaction, where interaction is understood through body movement, tangible manipulation, and digitally augmented physical space [19]. However, the artefact needed clear affordance and reliable feedback. If the book did not work immediately, participants became confused or disappointed. This shows that tangible interaction can create strong engagement, but only when the action and response are clear.

The spatial and atmospheric strategy changed how the exhibit felt. Curtains, lighting, and seating supported focus, calmness, and a more cinema-like atmosphere. The curtain separates the exhibit from the surrounding museum environment and made the space feel more intentional. However, it also created a threshold problem. Some visitors could interpret the curtain as a sign that the area was private or restricted. Therefore, spatial

framing needs to balance focus and openness. A welcome note or clearer entrance cue could help visitors understand that they are allowed to enter.

The temporal strategy changed how visitors understood the video's duration. The hourglass, progress bar, and timer showed that time awareness can be useful, especially when visitors decide whether to stay. Participants also mentioned that they wanted to know the length of the video before starting it, together with a short description of the content. They expected the video not to be too long, for example not around half an hour, and some suggested that a few minutes would be more suitable in the museum setting. This connects with Serrell's point that watching a museum video from beginning to end requires specific time commitment from visitors, which can be demanding within their limited time budget [34]. Serrell also highlights the importance of providing information such as running time and a brief content description and uses attracting power and holding power to understand how many visitors are attracted to videos and how long they stay watching them [34]. However, temporal cues can also distract if they are unclear or too visually dominant. In this study, temporal support worked best as a peripheral feature. Visitors did not need to look at it constantly, but it remained available when they wanted to check the remaining time.

Together, these strategies show that embodied engagement is not one single action. It is built through several small moments: being attracted, entering, acting, focusing, and staying.

6.4 Prototype-based knowledge from Research through Design

The book artefact revealed the importance of affordance, permission to touch, immediate feedback, and material fit. The hourglass and progress bar showed that temporal framing should remain peripheral to avoid distraction. The curtain showed that physical framing can support focus, but needs suitable material, coverage, and a clear invitation to enter.

The integrated prototype revealed problems that were not visible when the strategies were tested separately. For example, the numbered/button-like labels inside the book confused participants because they expected to press something. Replacing numbers with arrows made the sequence clearer. Similarly, the progress bar was replaced with a remaining-time timer because it became distracting. These changes provide evidence of prototype-based learning.

In this project, design knowledge emerged not only from what worked, but also from confusion, hesitation, technical failure, and reinterpretation. This fits with Research through Design because this approach actively engages in design and making rather than studying design from a distance [30]. Redström also describes Research through Design through practice, making, and judgement, which supports the idea that knowledge can emerge through situated design activity and reflection [30].

6.5 Design Implications

Several design implications emerged from the full design process, especially staff discussions and workshop activities, prototype testing, visitor feedback, observations and design iteration reflections. They are therefore presented as situated insights from one screen-based museum exhibit, rather than as general design rules. While Section 6.4 discusses prototype-based knowledge from the Research through Design process, this section summarizes what the findings suggest for future design of embodied augmentation in similar screen-based museum settings.

Table 2 : Design Insights Identified

Design area	Implication	Explanation for future design
Tangible trigger	Familiar artefacts can invite action, but their purpose must be clear	The book artefact showed that a physical object can work as an entry point into a video experience. Visitors do not need to keep focusing on it after triggering the video. However, the artefact should feel complete, intentional, and suitable for the exhibition topic.
Action and feedback	Tangible interaction needs an immediate and reliable response	When visitors open, touch, or move an object, the system should respond quickly and visibly. Delayed or unclear, visitors may become unsure whether they triggered the video correctly.
Spatial framing	Spatial enclosure should support focus without making the space feel closed	Curtains and lighting can make an existing video exhibit feel more intentional and focused. However, curtain material, color, thickness, coverage, and openness affect whether visitors feel invited or restricted. Welcome message or visible entrance can help communicate permission to enter.
Lighting	Lighting should be synchronized with the video experience	Lighting changes can increase attention and atmosphere, but they should respond with the video rather than feel separate. Light color should be chosen carefully because different colors create different emotional meanings.
Temporal framing	Time cues should be available but peripheral	Visitors may want to know the video length before deciding to stay. However, temporal cues can distract if they are too central, unclear, or visually active. A simple remaining-time display or short duration message may work better than a progress bar or separate object that visitors must interpret.
Interaction sequence	Embodied augmentation should be designed as one coherent sequence	Tangible, spatial, and temporal elements should guide visitors through the experience: what to do, what happens next, where to look, and how long it will take. The elements should not feel like separate additions.
Museum use	The setup should be flexible for different video topics	The tangible trigger does not always need to be a book. The same interaction logic can remain while the artefact, cover, or visual details change according to the video topic and exhibition aesthetics. Lighting could be adjusted to reflect the content and atmosphere of each video.

Together, these implications show how small tangible, spatial, and temporal additions can support engagement with an existing screen-based exhibit without replacing it.

6.6 Collaboration with the Museum as Part of the Design Process

The research question was formulated individually, but the research was conducted in close collaboration with the museum. Museum staff helped the researcher understand the setting, visitor behavior, practical constraints, and access to the exhibit space. This mattered because the study focused on augmenting an existing exhibit rather than designing a new one. The museum technician also supported technical decisions, material choices, installation safety, and prototype testing.

This collaboration showed that the research depended not only on design ideas, but also on the relationship between the researcher and museum stakeholders. Spending time in the museum and learning from staff helped reduce the distance between the researcher and the context before proposing solutions.

For future museum HCI research, staff collaboration should not be treated only as logistical support. Museum staff hold practical knowledge about visitors, space, maintenance, safety, and institutional values. Working closely with them can help researchers design interventions that are both interesting as prototypes and realistic within the museum setting.

6.7 Limitations

This study has several methodological limitations. The number of participants was limited, especially in strategy-specific and integration testing. Therefore, the findings are not statistically generalizable but should be understood as situated design insights from a small number of interactions in a museum setting.

Many participants were museum staff or university students. Although they were adults and could represent the target group of this exhibit, their feedback may not fully represent ordinary museum visitors, especially because some had prior knowledge of the museum or interaction design.

The study was conducted in one museum and around one video exhibit. The findings are shaped by that museum's spatial layout, visitor flow, values, and technical infrastructure, so they should be adapted carefully in other contexts.

The prototypes were temporary research artefacts, shaped by time, access, safety, materials, and technical compatibility. They should not be understood as final museum-ready installations.

Because the video content changes periodically, the study focused on the interaction format around the video rather than one specific documentary. Therefore, the findings mainly address embodied augmentation of the exhibit experience.

6.8 Future Work

Future work could test the integrated setup with more visitors over a longer period. This would help understand whether the design continues to attract engagement after the novelty effect disappears. Future studies could also explore other tangible triggers, such as connecting books or artefacts to different videos, or adapting the artefact to fit the topic and exhibition aesthetics. Further work could test alternative spatial and temporal framing, including lighting, enclosure, seating, and time cues. Since this study did not focus on age-group differences, future research could examine how children, adults, and families respond to these strategies. Finally, the augmentation approach could be tested in other museum contexts to understand how well the design insights transfer beyond this exhibit.

7 CONCLUSION

This study investigated issues in visitor engagement with a screen-based video exhibit, explored embodied augmentation strategies, and identified interventions for augmenting video exhibits without replacing them. The prototypes were developed iteratively, and each step shaped by earlier observation, testing, and reflection.

The findings show that each strategy reshaped visitor engagement differently. Tangible interaction created a more active starting point for the video experience, while spatial and atmospheric changes helped frame the

exhibit as a more focused and intentional space. Temporal framing made the video duration more visible but worked better as a peripheral support than as the focus of interaction.

The combination of strategies helped enhance the existing exhibit without making the interventions compete for the same function. Based on the data and analysis, tangible and spatial strategies created more immediate changes in visitor attention and behavior, while temporal framing supported the experience more subtly. Overall, this thesis shows that existing video exhibits can be meaningfully augmented through small material, spatial, and temporal interventions that work with the current exhibit rather than replacing it.

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AI Declaration

During the preparation of this thesis, ChatGPT was used only as a supplementary tool for grammar checking, language refinement, clearer wording, and suggestions related to the structure of some sections. The thesis itself was written by me. The research design, data collection, analysis, interpretation, and final decisions are my own.

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A APPENDICES

A.1 Consent form for strategy testing

A.2 Initial Observation Summary

A.3 Workshop data analysis summary

A.4 Strategy testing interview questions

A.5 Integrated testing interview questions

A.6 Sample feedback from strategy testing (Book-Artefact)

A.7 Strategy feedback form responses

A.8 Tangible Interaction – Technical Development

A.9 Tangible Interaction – High level circuit diagram

A.10 Integrated prototype Iteration 2 Design Changes

A.11 Sample Staff discussion data

A.12 Familiarization data sample

A.13 Design Ideation and sketches Sample

A.14 Categorize Design Ideas into Three Categories

A.1 Consent form for strategy testing

Consent Form for Participation in Thesis Study

Title of study:

Embodied Augmentation of Existing Video Exhibits: Reshaping Visitor Engagement Without Replacing the Screen

You are invited to participate in a study about the testing and evaluation of design ideas/prototypes developed for an existing museum exhibit. This study is being carried out as part of my Master's thesis in **Human Computer Interaction** at **Uppsala University**.

The purpose of this study is to explore how different design strategies and prototype interventions may influence visitor engagement and interaction with the exhibit. As part of this study, I am conducting testing sessions with students and museum staff.

What participation involves

If you agree to participate, you may be asked to:

- Interact with one or more prototype ideas or exhibit interventions
- Share your thoughts, opinions, and experiences during or after the session
- Complete a feedback form
- Allow the researcher to take notes during the session

Your participation is voluntary, and you may stop participating at any time without giving a reason.

Personal data and data processing

The purpose of processing personal data is to document and analyse participant feedback and interactions related to the prototype testing sessions.

The personal data that may be collected from you includes:

- Your spoken feedback during the session
- Written responses in feedback forms
- Observational notes about your interaction with the prototype/exhibit
- Audio recordings, video recordings, and/or photographs, **only if you separately consent below**

Your personal data will only be used for this thesis study and will be handled in accordance with the General Data Protection Regulation (GDPR). The collected data will be anonymized and stored securely and only accessed by the student researcher and supervisor. Personal data will be kept until the thesis has been examined and approved, after which identifiable material will be deleted.

If photographs, audio recordings, or video recordings are used, they will only be used for documentation and analysis within the thesis unless otherwise explicitly stated.

Other information about how Uppsala University processes personal data and your rights under GDPR can be found at:
https://www.uu.se/en/about-uu/data-protection-policy/

If you have questions or complaints, you can contact the Swedish Authority for Privacy Protection or Uppsala University's Data Protection Officer via:
dataskyddsombud@uu.se

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Consent

Please tick the boxes below if you agree:

[ ] I have received information about the study and understand what my participation involves.

[ ] I understand that my participation is voluntary and that I can withdraw at any time without giving a reason.

[ ] I consent to participate in this study.

Optional consent for documentation

Please tick separately for the following:

[ ] I consent to **audio recording** during the session.

[ ] I consent to **video recording** during the session.

[ ] I consent to **photographs being taken** during the session.

Signature: _____

Name: _____

Place and date: _____

A.2 Initial Observation Summary

Overall Summary of Observation Session

Observation period: 1:45 pm - 3:50 pm

Date: 03/02/2026

Method: Non-participant, in-situ naturalistic observation

All observations

Location: Museum video exhibit

Crowd level: Mostly low-medium

General Engagement Pattern

Across the observation session:

- Many visitors did not stop at all
- Those who stopped stayed 10-30 seconds
- Very few watched beyond 2-3 minutes
- No visitor watched the full documentary
- Children frequently exited first

The exhibit was repeatedly described as:

- "Not interactive"
- "Lengthy"
- "Not kids friendly"
- "Documentary type"

The video exhibit functioned primarily as a passive viewing station, and engagement was fragile and short-lived.

Behavioral Patterns Identified

Immediate Bypass Behavior

Several visitors:

- Walked past without turning their heads
- Identified the space as "library"
- Moved directly to aquarium/play areas
- Did not notice the video at all

I documented:

"By pass the place and just move to aquarium area without even looking at the video exhibit."

Interpretation

The exhibit does not generate:

- Peripheral attraction
- Sensory invitation
- Spatial interruption
- Social magnetism

It fails at the threshold moment - the first 3 seconds of encounter.

Short-Lived Curiosity

Some visitors:

- Pressed the button
- Sat down briefly
- Watched 10-30 seconds

- Left quickly

In one case:

- A child said “titta på video”
- Returned multiple times
- Stayed 2-3 minutes
- Eventually left

All observations

Interpretation

- The exhibit can generate initial curiosity, especially when:
- A child presses the button
- A parent invites participation
- The space is visually semi-covered

However:

Curiosity does not convert into sustained engagement.

Children’s Behavior

Recurring pattern:

- Children enter first
- Exit within seconds
- Prefer play area next door
- Show movement-based behavior (standing, wandering)

My reflection states:

- “Not kids friendly”
- “Visitors need something instant.”

All observations

Interpretation

The exhibit:

- Requires sustained attention
- Rewards cognitive focus
- Does not reward bodily engagement
- There is a mismatch between:
- Children's embodied energy
- The exhibit's passive format

This directly supports my embodied interaction framing.

A.3 Workshop data analysis summary

Theme	Example Staff Insight	Design Implication
Uncertainty about video duration	"When you watch a movie, you don't know when it's going to stop, so you just move away."	Make time visible to support visitor commitment
Temporal awareness	"When you watch a YouTube clip you look at where you are in the reel."	Physical time indicators (hourglass concept)
Button-pressing behavior	Visitors push buttons repeatedly and leave	Current interface encourages impulsive interaction
Embodied interaction	"If I have to work to keep the movie playing, I pay more attention."	Physical or effort-based triggers
Distraction concerns	Hourglass may attract more attention than the video	Balance engagement with content focus
Audience mismatch	"We get a lot of kids here even though it's not for them."	Design cues should signal intended audience
Environmental design	Curtains, lighting, spatial separation suggested	Spatial framing can guide behavior
Architecture vs rules	"We need architecture to do the talking."	Behavioral cues embedded in space

A.4 Strategy testing interview questions

General questions

1. What did you notice first when you approached the exhibit?
2. What did you think you were supposed to do?
3. Was anything unclear or confusing?
4. How did this change your experience of the video?
5. Did it affect how long you wanted to stay or pay attention?
6. How did it feel to interact with it physically?
7. What worked well for you?
8. What would you change?

Temporal framing / hourglass

1. What did you think the hourglass was meant to do?
2. Did it help you understand or feel the duration of the video?
3. Did it make you more willing to stay with the video?
4. Did it feel connected to the exhibit, or separate from it?
5. How did using the hourglass affect your attention?
6. Did it distract you from the video?
7. How would you like to make time visible in this exhibit?

8. Do you think time becomes visible can improve your experience?

Spatial framing

1. What did you notice about this space?
2. How did the curtain or lighting affect the way you approached the exhibit?
3. Did the space feel more inviting, focused, or different in some way?
4. Did the changes help you pay more attention to the exhibit or video?
5. Did anything in the space feel unclear, awkward, or uncomfortable?
6. Did it affect the duration you stay?
7. What kind of atmosphere did the space create for you?
8. Do you think these changes improve the exhibit? Why or why not?

Book-specific questions

1. What did you think the book was for?
2. Did it feel natural to open the book to start the video?
3. Did the prompt help you understand what to do?
4. Did opening the book make you feel more involved in the exhibit?
5. Did the shape of the book help you understand the interaction?
6. Did holding or opening the book affect your attention to the video?
7. Did anything feel awkward or unusual about this interaction?
8. Best short version for real session

A.5 Integrated testing interview questions

Core Questions

1. What did you notice first when you approached the exhibit?
2. What did you think you were supposed to do?
3. Was the sequence from opening the book to starting the video clear?
4. How did the display messages affect your understanding of the interaction?
5. Did the countdown help prepare you for the video?
6. Did the progress bar help during the video?
7. How did the curtain and lighting affect your experience?
8. Did the whole setup feel like one coherent experience?
9. Was anything confusing, awkward, or unnecessary?
10. What would you change before testing with visitors?

Optional Staff-Specific Questions

1. Do you think this setup fits the exhibit and museum context?
2. Does it improve the exhibit without taking attention away from the content?
3. Do you think visitors would understand what to do without help?
4. Which part needs the most refinement before final testing?

A.6 Sample feedback from strategy testing (Book-Artefact)

Participant Feedback Form

Participant ID:

Date:

Please tick one answer for each statement.

Statement	Strongly agree	Agree	Not sure	Disagree	Strongly disagree
I noticed the book artifact easily.					
The prompt was easy to understand.					
It felt natural to open the book to start the video.					
The book made the exhibit more interesting.					
The book helped me feel more involved in the exhibit					
The interaction was clear and easy to use.					

A.7 Strategy feedback form responses

Statement	**Participant 1**	**Participant 2**	**Participant 3**	**Participant 4**	**Participant 5**
Hourglass					
It was easy to understand what to do.	disagree	disagree	disagree	disagree	agree
This made the exhibit more engaging.	agree	not sure	disagree	disagree	agree
This helped me pay attention to the video.	disagree	disagree	disagree	disagree	not sure
The interaction felt natural.	strongly disagree	not sure	strongly disagree	strongly disagree	strongly agree
I would like to use this in a museum.	not sure	not sure	not sure	not sure	agree
I felt the time.	strongly agree	agree	agree	agree	not sure
Spatial.					
I noticed the changes in the exhibit space.	strongly agree	agree	disagree	not sure	agree
This made the exhibit	strongly agree	not sure	disagree	agree	strongly agree

more engaging.					
The curtain and/or lighting helped me focus more.	strongly agree	agree	disagree	strongly agree	strongly agree
The space felt clear and easy to understand.	agree	agree	disagree	not sure	agree
The changes made the exhibit feel more engaging.	strongly agree	agree	disagree	strongly agree	strongly agree
I would like to use this in a museum.	strongly agree	agree	disagree	strongly agree	strongly agree
Artefact					
I noticed the book artefact easily.	agree	not sure	disagree	strongly agree	strongly disagree
The prompt was easy to understand.	strongly agree	strongly agree	strongly agree	strongly agree	strongly agree
It felt natural to open the book to start the video.	strongly agree	strongly agree	agree	strongly agree	strongly agree
The book made the exhibit more interesting.	strongly agree	strongly agree	strongly agree	agree	strongly agree
The book helped me feel more involved in the exhibit.	not sure	strongly agree	strongly agree	agree	not sure
The interaction was clear and easy to use.	strongly agree	agree	agree	agree	strongly agree

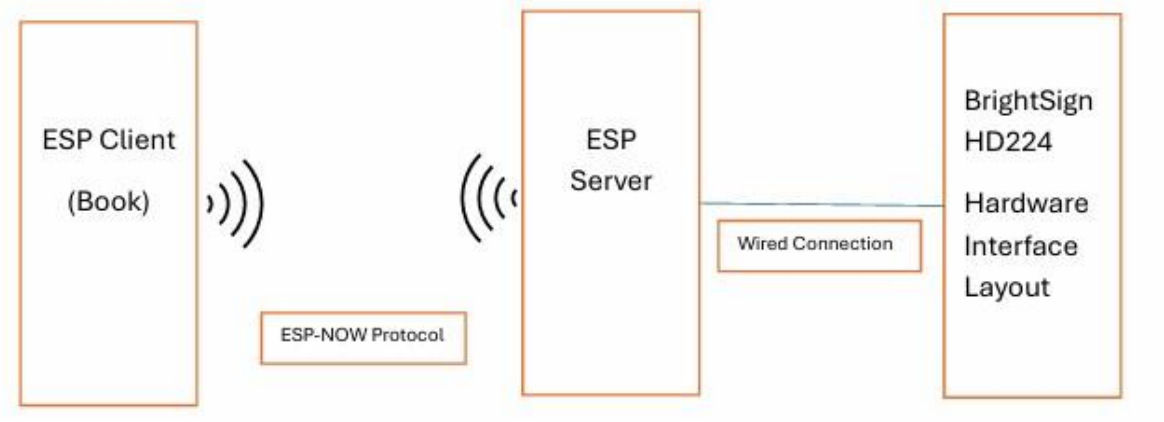
A.8 Tangible Interaction – Technical Development

This interaction aimed to transform passive viewing into a more embodied form of engagement, where visitors' physical action directly influenced the digital content. Technically, a hall-effect mechanism [1] was used, with a hall-effect sensor [1] and magnet detecting whether the book was open or closed. After a Tinkercad [39] simulation, early tests were carried out using a simple Arduino-based circuit [2], which was later developed into a wireless ESP32-based setup to make the prototype more stable and suitable for the museum environment.

The prototype was tested with the museum technician to check compatibility with the existing museum electronics and infrastructure. This testing showed that a wired connection to the screen's GPIO input [16] was not ideal because of distance, voltage drop, and the risk of disconnection through repeated handling. Based on this, the circuit was modified in the next iteration using an ESP32 [32] based client and server module, communicating through the ESP-NOW protocol [31]. This created a more stable wireless connection, reduced the

need for long wires, and allowed the components to be placed inside the book artefact. Expert support was used for soldering and improving the circuit to make the prototype more reliable during testing (see figure 1).

A.9 Tangible Interaction – High level circuit diagram

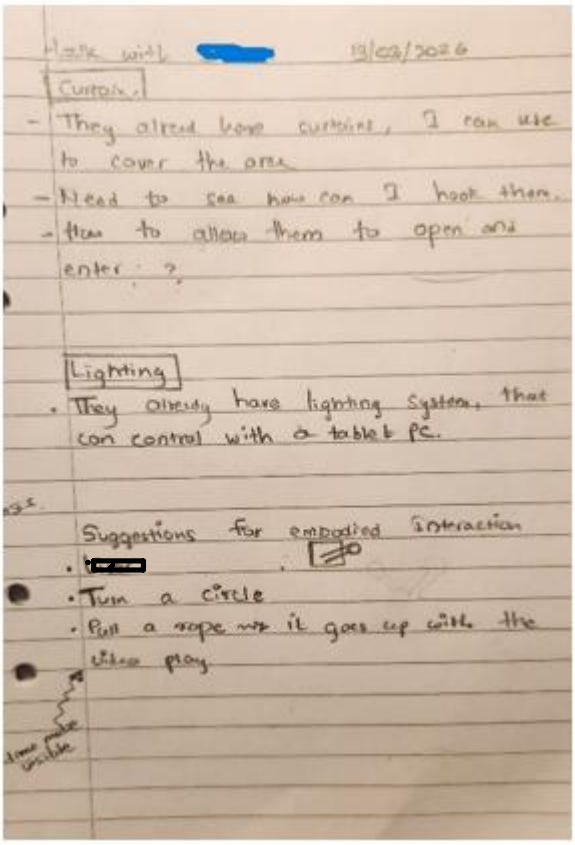


A.10 Integrated prototype Iteration 2 Design Changes

The second integrated prototype was developed based on feedback from the first integration test. The progress bar on the small display was replaced with a remaining-time display because it was seen as clearer and less distracting. As one participant said, "*I would like that more than a percentage.*" The button-shaped numbers inside the book were also replaced with arrows, following the suggestion that "*you could replace it with arrows*." The timing between the messages and the countdown was adjusted because one participant felt that "*maybe the countdown time was a bit too long*." The red lighting was also changed to a warmer yellowish light to better fit the museum atmosphere.

The book topic was changed from "Inside the Pond" to "Frog Life Cycle" to make the connection between the artefact and the video content clearer. This followed one participant's suggestion that the title could refer more specifically to pond creatures, such as "*small insects living in the pond*." No other major changes were made to the setup.

A.11 Sample Staff discussion data



Talk with [redacted] 13/03/2026

Curtain:

- They already have curtains, I can use to cover the area
- Need to see how can I hook them.
- How to allow them to open and enter ?

Lighting

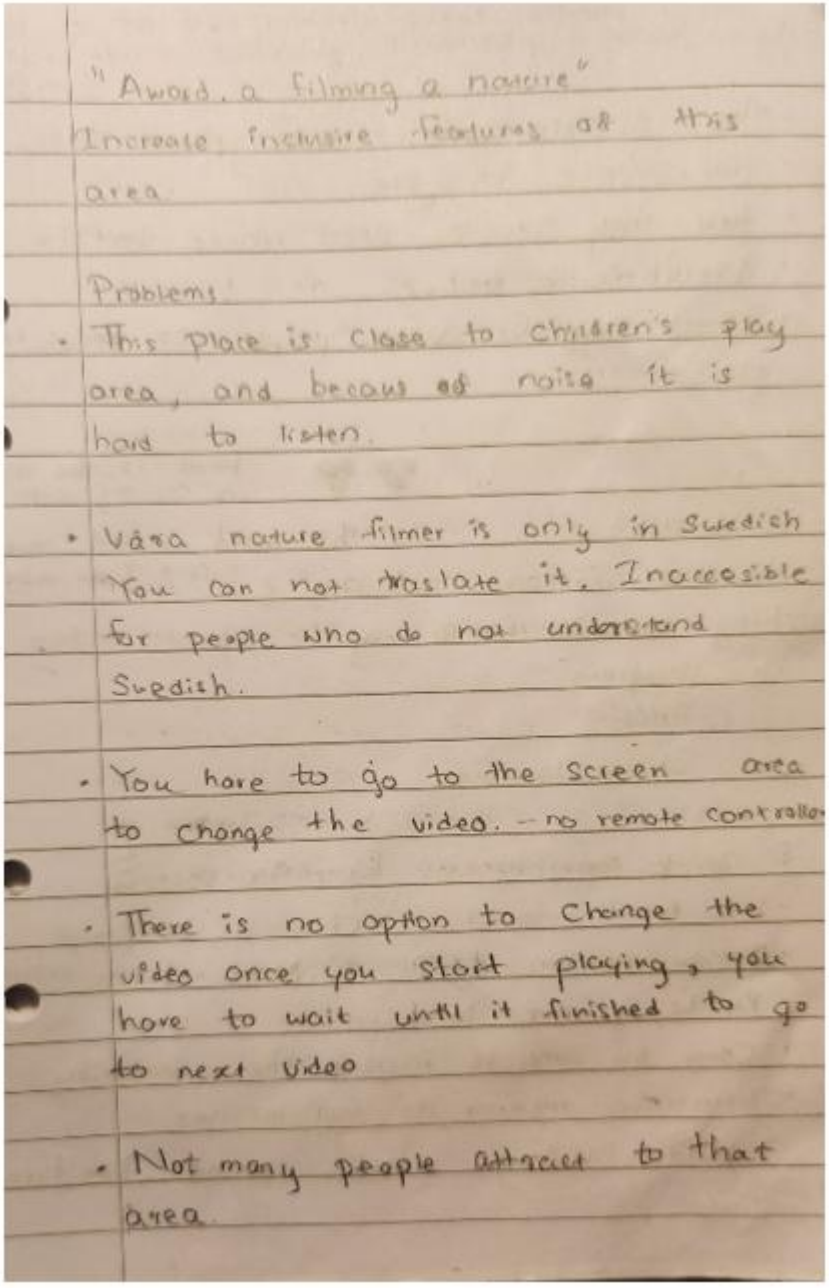
- They already have lighting system, that can control with a tablet PC.

Suggestions for embodied interaction

- [redacted]
- Turn a circle
- Pull a rope we it goes up with the video play

time make visible

A.12 Familiarization data sample

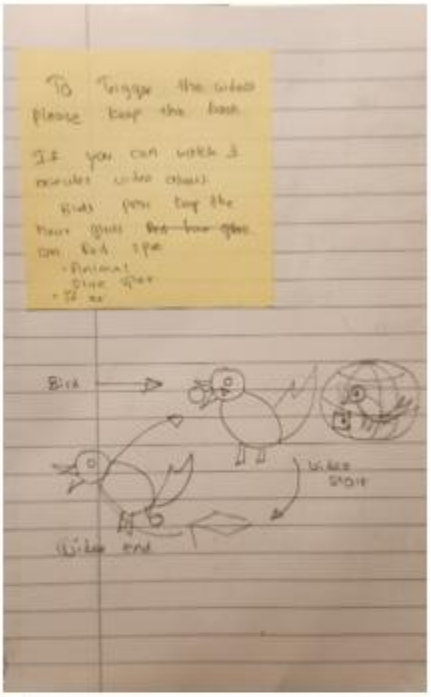


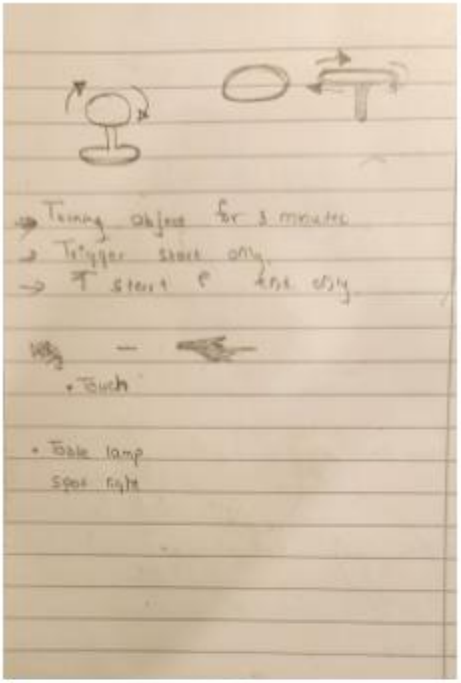
"Award. a filming a nature"
Increase inclusive features at this area

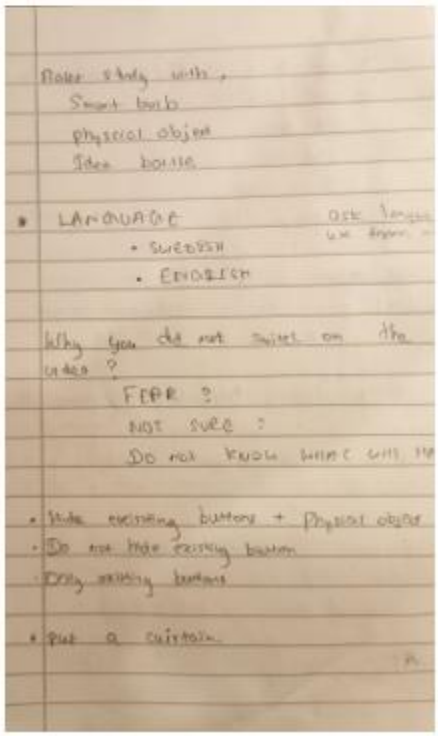
Problems:

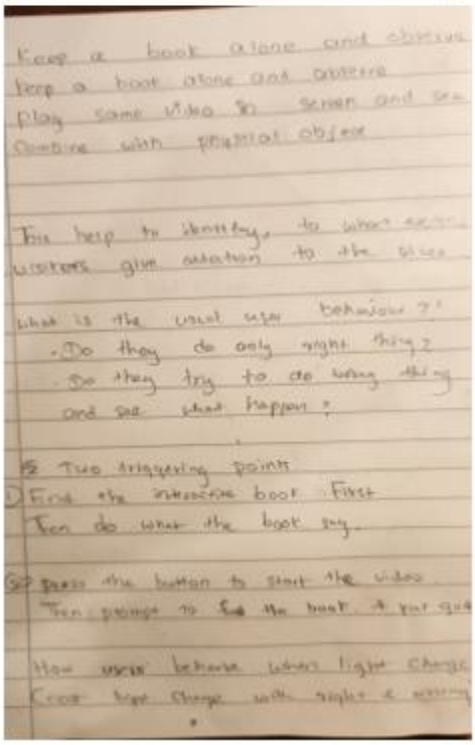
- This place is close to children's play area, and because of noise it is hard to listen.
- Vårsa nature filmer is only in Swedish You can not traslate it. Inaccesible for people who do not understand Swedish.
- You have to go to the screen area to change the video. - no remote controller
- There is no option to change the video once you start playing, you have to wait until it finished to go to next video
- Not many people attracet to that area

A.13 Design Ideation and sketches Sample









A.14 Categorize Design Ideas into Three Categories

